

Received power: 28 visual options

A figure storyboard for the integrated MH370 estimation paper

The visual story

The terminal did not measure position. It measured a signal after aircraft-spacecraft geometry, a directional antenna, propagation, channel effects and possible terminal power control had all acted on it. The figure sequence should make that conditional nature visible before showing the statistical result.



SUGGESTED READER JOURNEY



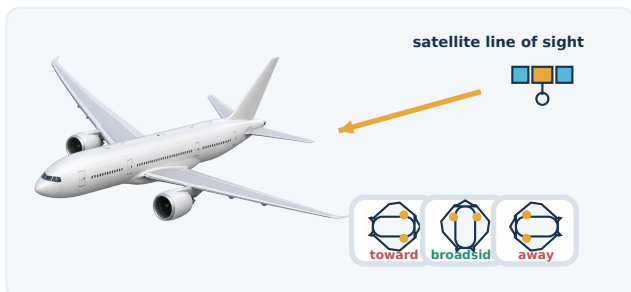
What is already data-anchored

MH371 phase headings and reconstructed gains: 01:55—236°T, 10.4 dBic; 03:21—158°T, 14.1 dBic; 03:29—about 219°T, 11.7 dBic. The same R1200-0-36E3 channel is used in the counterfactual options.

What remains deliberately schematic

The thumbnail lobes, attitude cases and posterior islands communicate visual form only. Selected plates will be regenerated from the full reconstructed gain surface, particle output and uncertainty model, with provenance stated in the caption.

01 GEOMETRY



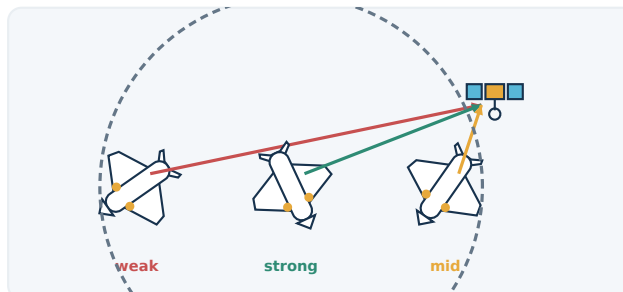
Perspective geometry triad

Toward, broadside and away at low satellite elevation, with the side-mounted arrays called out.

EXPLAIN + MODEL

Opening figure; immediate intuition

02 GEOMETRY



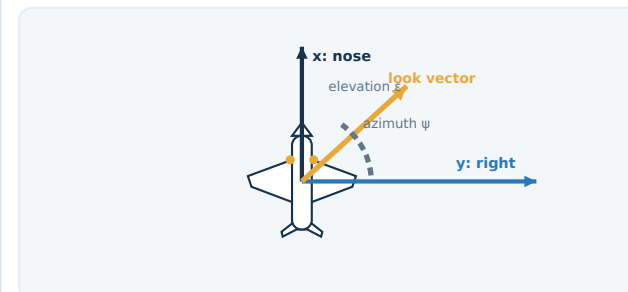
Rotate the aircraft, hold the satellite fixed

Three headings at one ground point isolate why heading—not position alone—changes antenna gain.

EXPLAIN

Opening alternative / animation

03 GEOMETRY



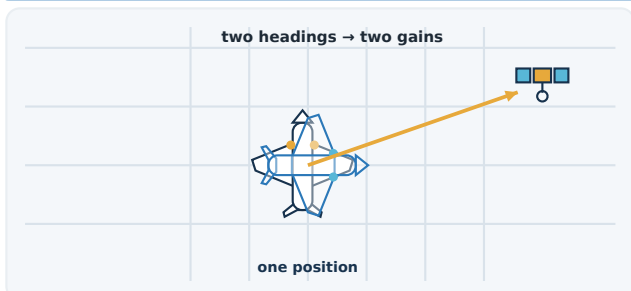
Aircraft-space coordinate frame

Defines relative azimuth and elevation, body axes, attitude and the satellite line-of-sight vector.

EXPLAIN + DEFINITIONS

Technical inset beside equations

04 GEOMETRY



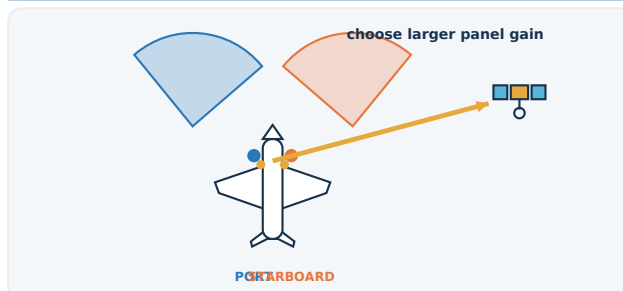
Same map position, different received power

A plan-view counterexample: identical aircraft and satellite positions, but different headings and gains.

EXPLAIN

General-reader bridge

05 GEOMETRY



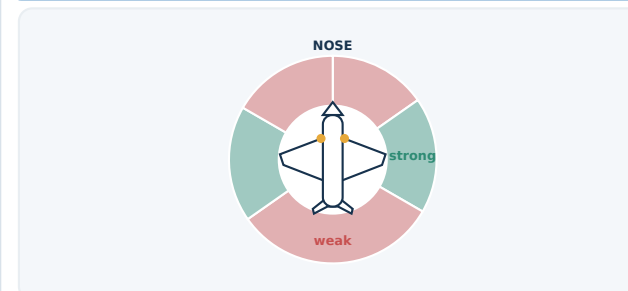
Port/starboard antenna hand-off

Shows the two canted side arrays and which panel supplies the larger gain as the look direction moves.

EXPLAIN + MODEL

Hardware/geometry explanation

06 GEOMETRY



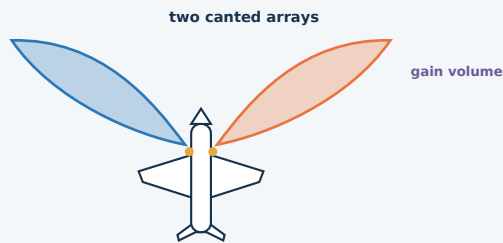
Horizon compass of strong and weak sectors

A top-down coverage ring makes the low-elevation nose/tail weakness visible without a 3-D surface.

MODEL

Compact main-text alternative

07 GAIN

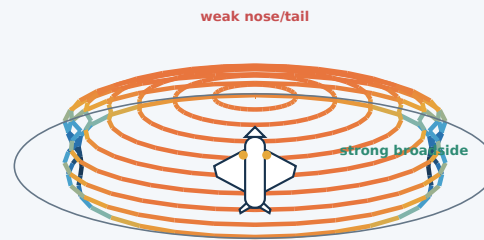
**Twin gain lobes mounted on the airframe**

An aircraft carries the two directional gain volumes, connecting the side-mounted hardware to the pattern.

EXPLAIN + MODEL

Hero alternative / graphical abstract

08 GAIN

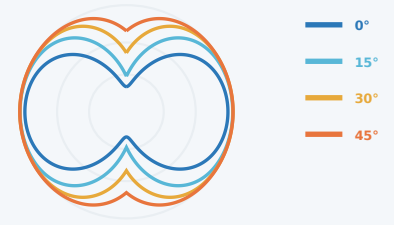
**3-D gain hemisphere**

A colour-coded upper-hemisphere surface shows broadside maxima and low-elevation nose/tail valleys.

MODEL

Primary gain-pattern figure

09 GAIN

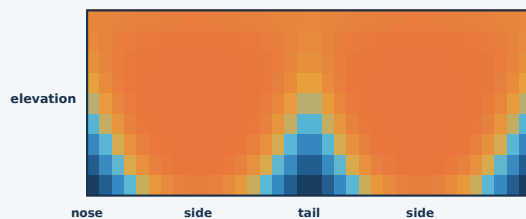
**Polar slices at several elevations**

Azimuthal gain curves at 0°, 15°, 30° and 45° elevation improve the dissertation's two-slice plot.

MODEL + DATA

Quantitative main or supplement

10 GAIN

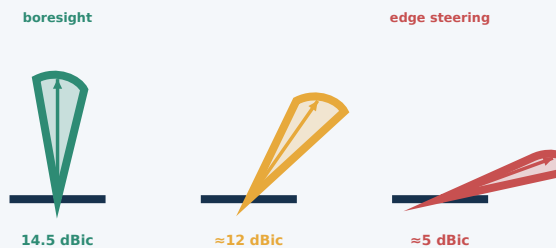
**Elevation-azimuth gain heat map**

A rectangular aircraft-space map is easiest for readers to query and for particles to interpolate.

MODEL + DATA

Methods figure / model validation

11 GAIN

**Beam steering and scan loss**

A three-step diagram follows the beam from boresight toward the array edge as peak gain falls.

EXPLAIN

Phased-array explainer

12 GAIN

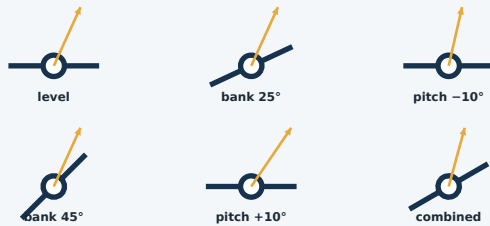
**Bore-sight-to-horizon gain profile**

A single annotated curve turns the 3-D pattern into the one-dimensional quantity used in the link budget.

MODEL + DATA

Compact quantitative inset

13 ATTITUDE



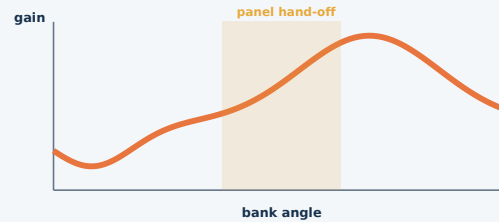
Pitch-bank-yaw coverage matrix

Small multiples show how the same spacecraft direction moves across the gain map under attitude changes.

MODEL

Controlled/unusual-flight discussion

14 ATTITUDE



Bank-angle sweep

A continuous sweep shows when a low-elevation spacecraft enters a strong lobe, weak sector or hand-off region.

MODEL

Sensitivity figure

15 ATTITUDE



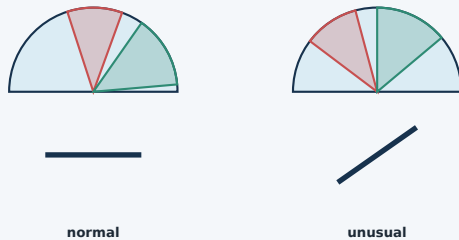
Climb and descent sweep

Contrasts level flight, descent and climb without conflating flight-path angle with aircraft-space elevation.

MODEL

Controlled-flight scenario inset

16 ATTITUDE



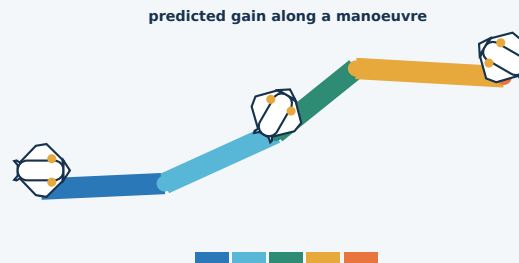
Normal versus unusual attitude sky domes

Two hemispheres compare routine cruise coverage with a steep bank/pitch case, including no-coverage regions.

MODEL

End-of-flight caveat / supplement

17 ATTITUDE



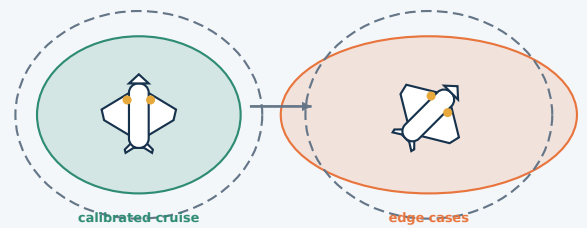
Coverage ribbon through a controlled turn

A route ribbon is coloured by predicted gain while heading and bank evolve through a turn.

MODEL + TRAJECTORY

Trajectory-model integration

18 ATTITUDE



Operational envelope and edge cases

A two-panel map separates well-constrained normal-attitude behaviour from extrapolative unusual attitudes.

MODEL + UNCERTAINTY

Uncertainty disclosure

19 MH371

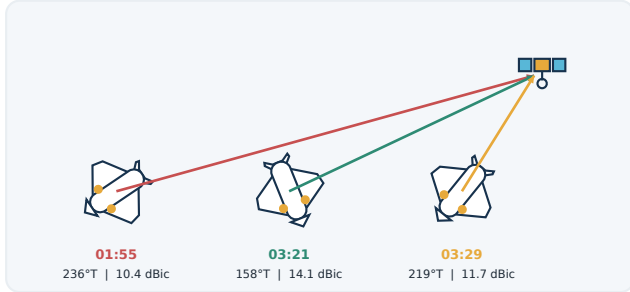


MH371 route as a natural experiment

The route map marks 01:55, 03:21 and 03:29, where heading changes produced large predicted gain changes.

DATA
Primary validation figure

20 MH371

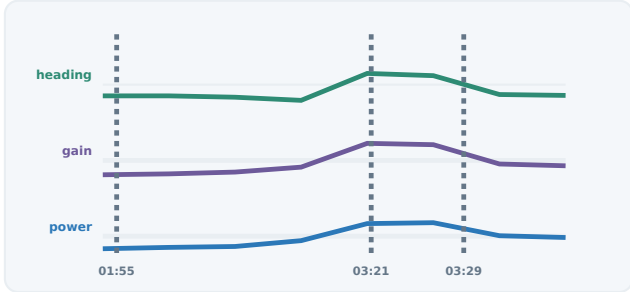


Three matched geometry snapshots

At each phase, a plane, line of sight, heading, predicted gain and path-corrected power are shown together.

DATA + MODEL
General-reader validation figure

21 MH371

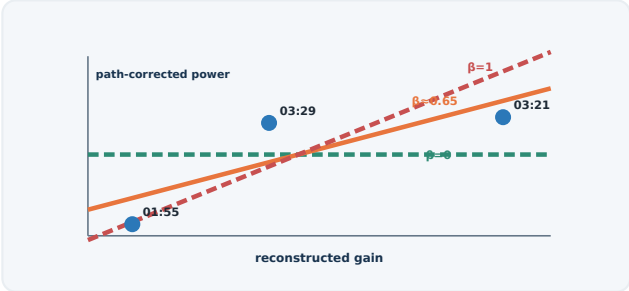


Time-aligned heading, gain and power

Stacked traces show whether received-power changes occur where the geometry model predicts them.

DATA + MODEL
Technical validation figure

22 MH371

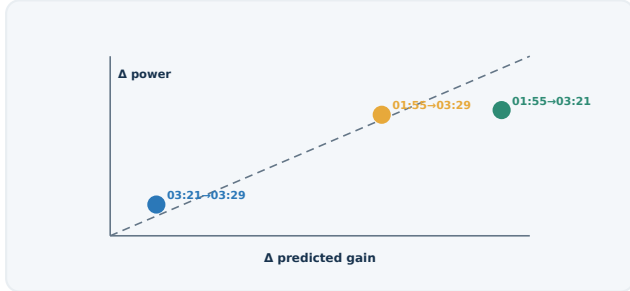


Same-channel counterfactual test

The three phase means are compared with complete, partial and zero gain-precompensation predictions.

DATA + INFERENCE
Core hypothesis-test figure

23 MH371



Observed change versus predicted gain change

A change-on-change plot removes the arbitrary power intercept and makes agreement or attenuation visible.

DATA + INFERENCE
Robustness / supplement

24 INFERENCE

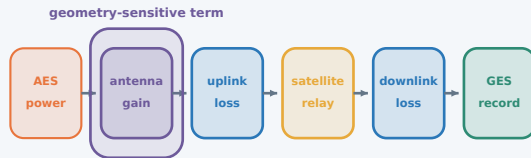


Gain-precompensation coefficient β

A visual continuum links $\beta=0$ (complete compensation), the estimate, and $\beta=1$ (none).

INFERENCE
Hypothesis-test summary

25 INFERENCE



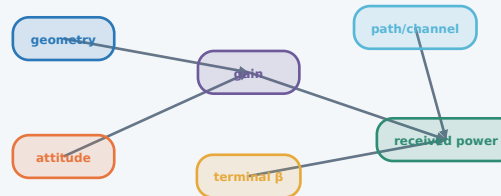
End-to-end received-power chain

A left-to-right link budget separates AES gain, space loss, satellite/ground terms and recorded power.

EXPLAIN + MODEL

Methods overview

26 INFERENCE



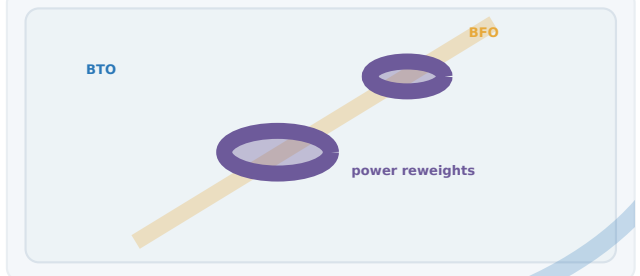
Causal graph and nuisance separation

Geometry drives gain; channel, path, terminal control and slower RF offsets also influence the observation.

INFERENCE + MODEL

Statistical-model architecture

27 INFERENCE



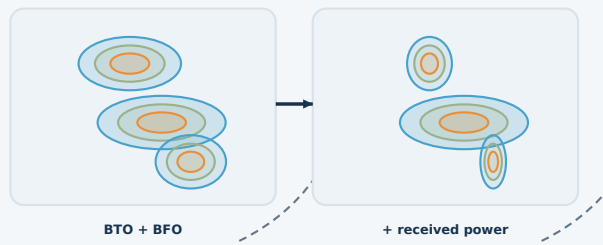
How power intersects the BTO/BFO solution

Layered likelihood ridges show power narrowing—but not independently locating—the satellite-constrained path.

INFERENCE

Integrated-estimator explanation

28 INFERENCE



Posterior before and after received power

Side-by-side 00:11 maps reveal which modes are reweighted when the received-power likelihood is added.

INFERENCE + DATA

Results figure